CIE AS PHYSICS (9702) • QUESTION PAPER

9.3 Resistance & Resistivity

Syllabus Section: 9. Electricity **Paper Type:** Structured Questions (Paper 2) **Total:** 11 Questions (106 Marks)

Question 1 Easy [9 marks]

Part (a) [1 mark]

A straight wire of unstretched length L has an electrical resistance R. The area of the cross-section A of the wire may be assumed to be constant.

State the relation between R, L, A and the resistivity ρ of the material of the wire.

Part (b) [3 marks]

Measurements made for a sample of metal wire are shown in Table 1.1.

Table 1.1

Quantity	Measurement
resistance	6.1 Ω
diameter	0.44 mm
length	1390 mm

State the most appropriate instrument to use to measure

(i) the resistance of the wire

[1]

(ii) the diameter of the wire

[1]

(iii) the length of the wire.

[1]

Part (c) [3 marks]

Show that the resistivity of the metal is 6.67×10^{-7} Ω m.

Part (d) [2 marks]

State and explain what would happen to the resistivity of the wire if the diameter was doubled.

Question 2 Easy [10 marks]

Part (a) [2 marks]

On Fig 1.1, sketch the change in resistance with temperature for a thermistor.

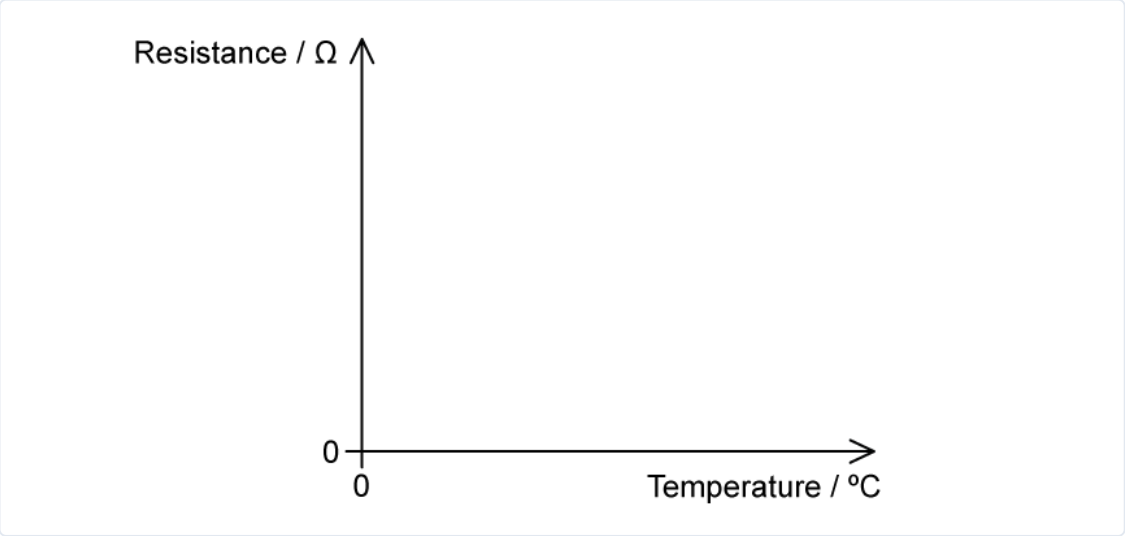


Fig 1.1

Part (b) [3 marks]

At 20 °C a wire is used to create a thermistor and has an electrical resistivity of 5.6×10^{-8} Ω m.

State and explain what would happen to the resistivity of the thermistor if its temperature is increased to 60 °C.

Part (c) [3 marks]

(i) State the meaning of *LDR*.

[1]

(ii) On Fig 1.2, sketch the change in resistance with light intensity for an LDR.

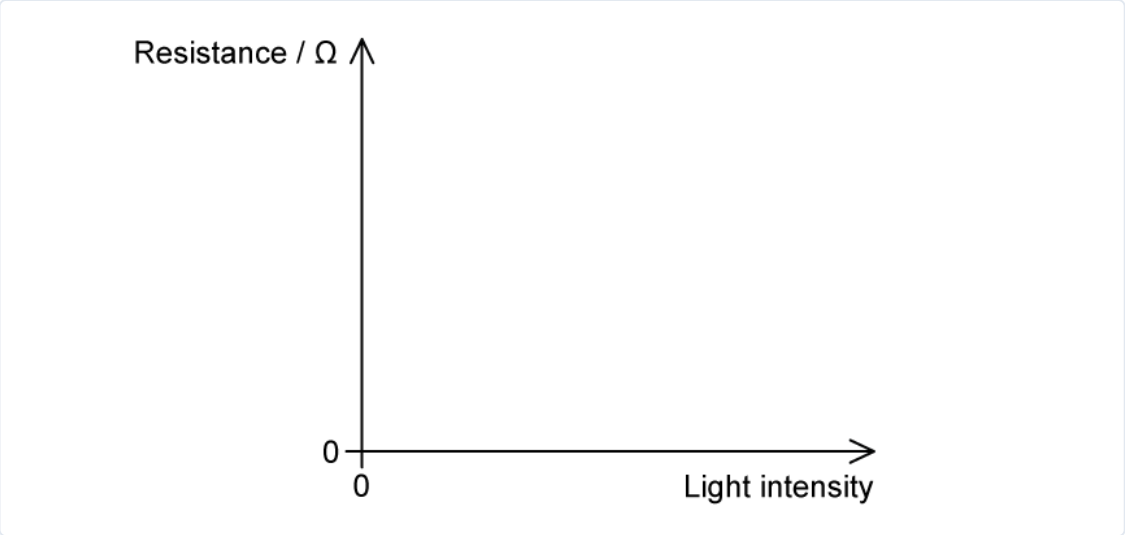


Fig 1.2

[2]

Part (d) **[2 marks]**

An LDR is connected to a circuit to operator an automatic street light. The resistance of the LDR is measured to be 5 kΩ at one time and 20 MΩ at another.

State which resistance the LDR is likely to be at during the day and which it will be at night.

Question 3 Easy [11 marks]

Part (a) [4 marks]

Complete Table 1.1 by placing a tick (✓) in the correct column to show whether each of the quantities affects or does not affect the resistance of a conducting wire.

Table 1.1

quantity	does not affect resistance	affects resistance
cross-sectional area		
length		
resistivity		
mass		

Part (b) [2 marks]

A lamp is designed to take a current of 0.65 A when connected across a 24 V mains supply.

The filament is made from one of the three metals from Table 1.2 with a diameter of 66 μm.

Table 1.2

Metal	Resistivity / Ω m
Nichrome	1.1×10^{-6}

Tungsten	5.5×10^{-7}
Copper	1.7×10^{-8}

Calculate the cross-sectional area of the metal wire.

Part (c) [2 marks]

Calculate the resistance of the wire.

Part (d) [3 marks]

Hence, or otherwise, determine which metal the filament is made from for a length of 23 cm.

Question 4 Medium [9 marks]

Part (a) [3 marks]

The photograph shows a statue of Buddha in Sri Lanka, which is protected by a lightning conductor.



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During a storm, a potential difference of 2.7 MV was generated between a cloud and the top of the lightning conductor on the statue. A flash of lightning passed between the cloud and the lightning conductor, producing a current of 25 kA for a time of 7.5 ms.

Calculate the energy transferred by the lightning strike.

Energy transferred = ...

Part (b) [3 marks]

The lightning conductor is made from copper.

Sketch an *I*–*V* graph for the rod when it conducts a lightning strike to earth

Part (c) **[3 marks]**

Explain any changes which the lightning rod would undergo during a lightning strike.

Question 5 Medium [8 marks]

Part (a) [3 marks]

Fig. 1.1. shows the graph of V against I for a non-ohmic conductor.

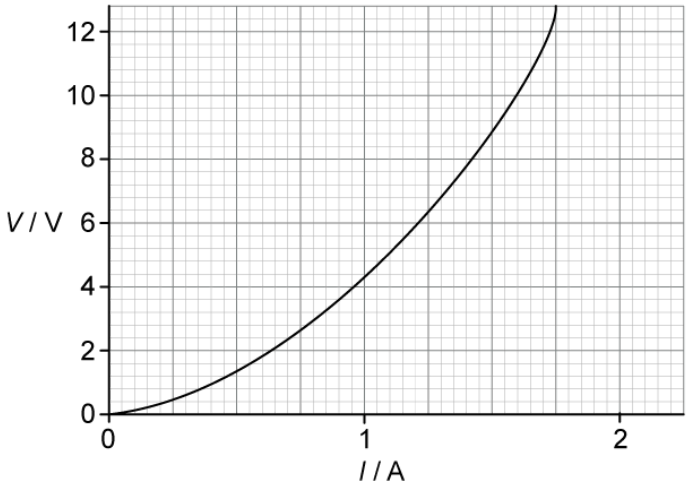


Fig. 1.1.

Calculate the maximum resistance of the lamp over the range shown by the graph.

Part (b) [2 marks]

Using the axes in Fig. 1.2. sketch the graph of current against potential difference for a diode.

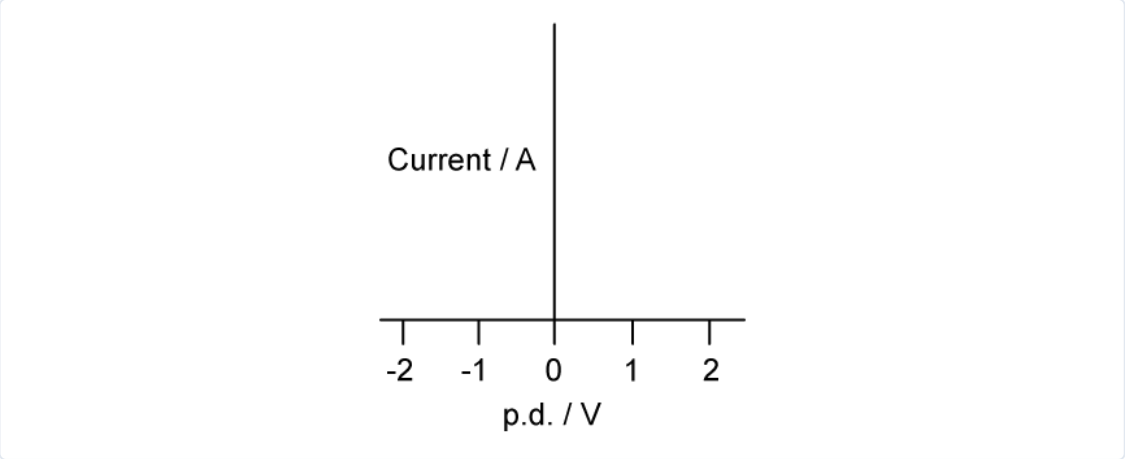


Fig. 1.2.

Part (c) [3 marks]

For the graph in Fig. 1.3. determine

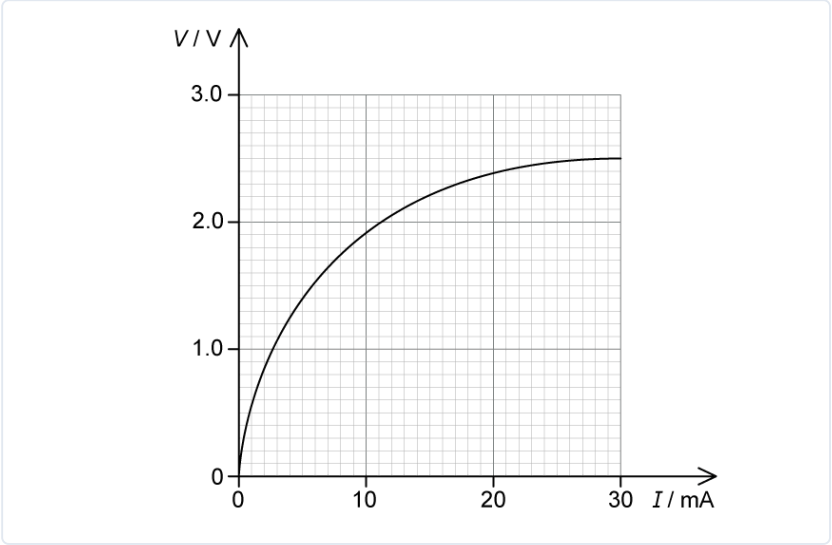


Fig. 1.3.

(i) the resistance when current = 10 mA

[2]

(ii) the power when current = 10 mA

[1]

Question 6 Medium [10 marks]

Part (a) [2 marks]

An electrical component has the I-V characteristic shown in Fig. 1.1.

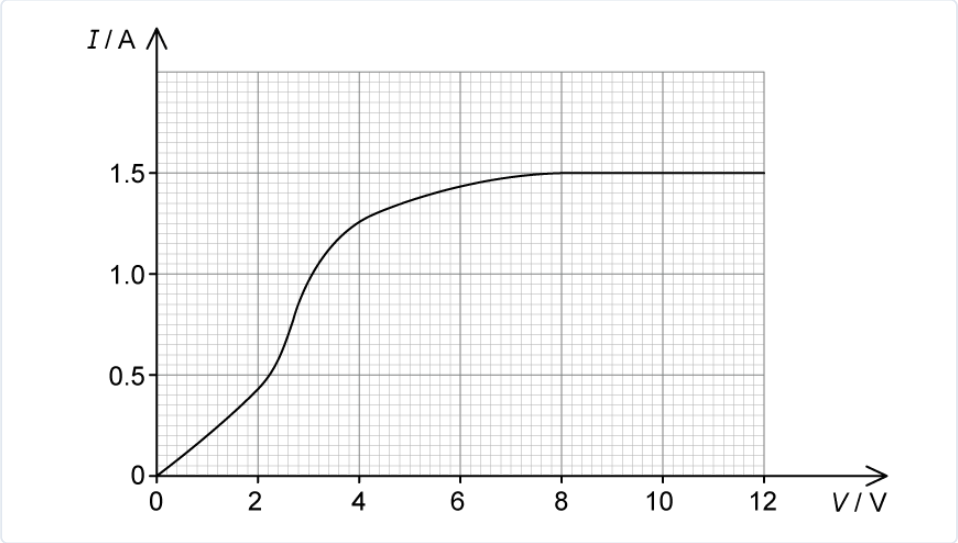


Fig. 1.1

Determine the resistance of the component when a potential difference of 6.0 V is applied across it.

Part (b) [2 marks]

For the component in part **(a)** calculate the minimum value of R as the potential difference across it changes over the range 0-10 V.

Part (c) [6 marks]

For the component in part **(a)** calculate

(i) The power dissipated when V = 2.0 V

[2]

(ii) The maximum power dissipated before the component heats up

[2]

(iii) The value of resistance at the point identified in part (ii)

[2]

Question 7 Medium [13 marks]

Part (a) [5 marks]

An electrical heating element, made from uniform nichrome wire, is required to dissipate 600 W when connected to the 230 V mains supply.

The radius of the wire is 0.16 mm.

Calculate the length of nichrome wire required.

Resistivity of nichrome = 1.1×10^{-6} Ω m

Part (b) [2 marks]

Suggest **two** properties that the nichrome wire must have to make it suitable as an electrical heating element.

Part (c) [3 marks]

Explain why the resistivity of the nichrome wire changes with temperature.

Part (d) **[3 marks]**

An engineer wants to use the same uniform nichrome wire to use in another identical electrical heating element which is also required to dissipate 600 W when connected to the 230 V mains supply.

The only nichrome wire they have available has a radius of 0.08 mm.

Calculate the length needed for this new nichrome wire to produce the same current through the heating element.

Question 8 Medium [9 marks]

Part (a) [3 marks]

A student investigates the resistivity ρ of a length of a wire l, with resistance R, diameter d and a potential difference V through it.

Determine the resistivity, in terms of ρ, of a wire of the same length with a diameter $2d$ and a potential difference $3V$ through it instead.

Part (b) [3 marks]

The student decides to investigate how the cross-sectional area of a wire affects its resistance. The circuit they use to carry out the experiment is shown in Fig 1.1.

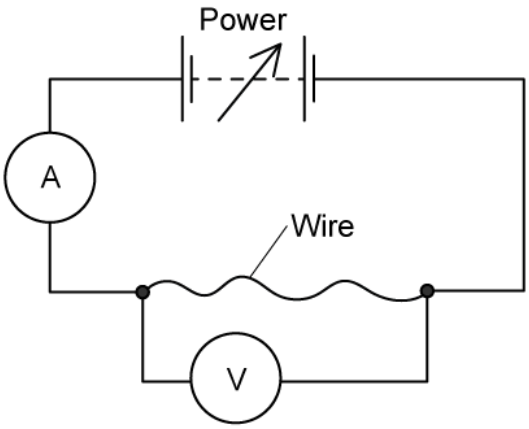


Fig 1.1

Describe a method to accurately determine the cross-sectional area of the wires.

Part (c) **[3 marks]**

The student uses three pieces of nichrome wire, each 1.5 m long. The cross-sectional area of the pieces of wire are

0.45 m^2	0.21 m^2	0.89 m^2

Sketch a graph of voltage *V* against the current *I* for each of the wires.

Question 9 Medium [7 marks]

Part (a) [4 marks]

At room temperature, a metal has a resistivity of 3.0×10^{-7} Ω m.

A 2.5 m length of wire made from this metal has a cross-sectional area of 0.50 mm^2.

Calculate the power dissipated in this length of wire when it carries a current of 10 mA.

Part (b) [2 marks]

State and explain the assumption that is necessary for calculating the answer to part (a).

Part (c) [1 mark]

In electrical power transmission conventional copper wires are often replaced with wires which conduct with wires with an effective resistance which is close to zero, called superconducting wires.

Explain why the superconducting wires increase the efficiency of electrical power transmission.

Question 10 Hard [7 marks]

Part (a) [2 marks]

A wire of cross-sectional area A is from made metal of resistivity ρ. The wire is extended. Assume that the volume V of the wire remains constant as it extended.

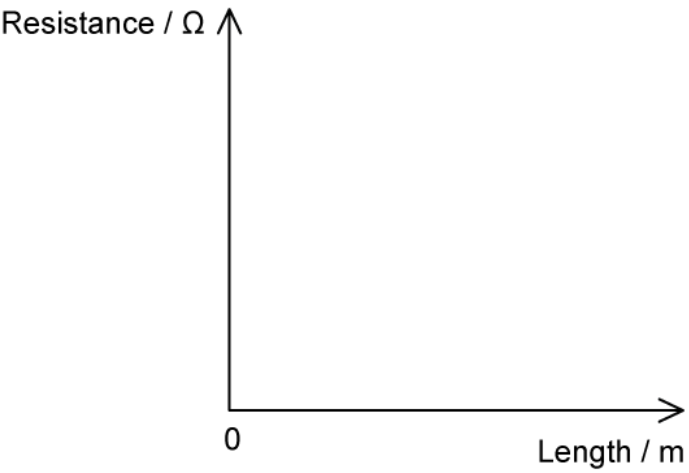


Fig 1.1

On Fig. 1.1, sketch the variation with length of the resistance as the wire is extended.

Part (b) [3 marks]

Two resistors X and Y have resistance R_X and R_Y respectively. The resistors are connected in series with a battery with electromotive force (e.m.f) E and zero internal resistance, as shown in Fig 1.1.

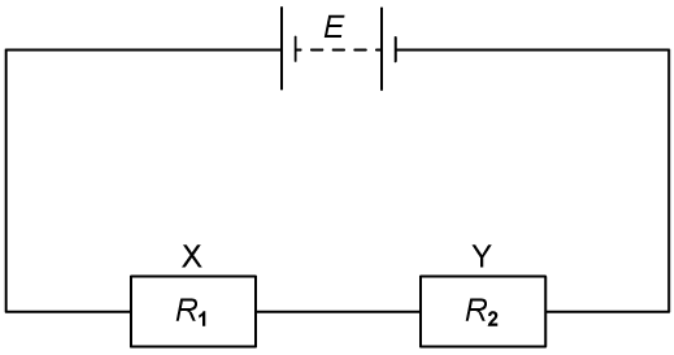


Fig 1.1

The resistors are made from metal wires. Data from the resistors are given in Table 1.1.

Table 1.1

resistor	X	Y
diameter of metal	$\frac{\rho}{3}$	ρ
length of metal	$2l$	l
resistivity of metal	$4d$	d

Use information from Table 1.1 to determine the ratio

$$\frac{\text{power dissipated in Y}}{\text{power dissipated in X}}$$

Part (c) [2 marks]

Show that the product of the ratio $\frac{\text{power dissipated in Y}}{\text{power dissipated in X}}$ of X and Y connected in series and connected in parallel is 1.

Question 11 Hard [13 marks]

Part (a) [5 marks]

Thin films of carbon are sometimes used in electronic systems. Typical dimensions of such a film are shown in Fig. 1.1.

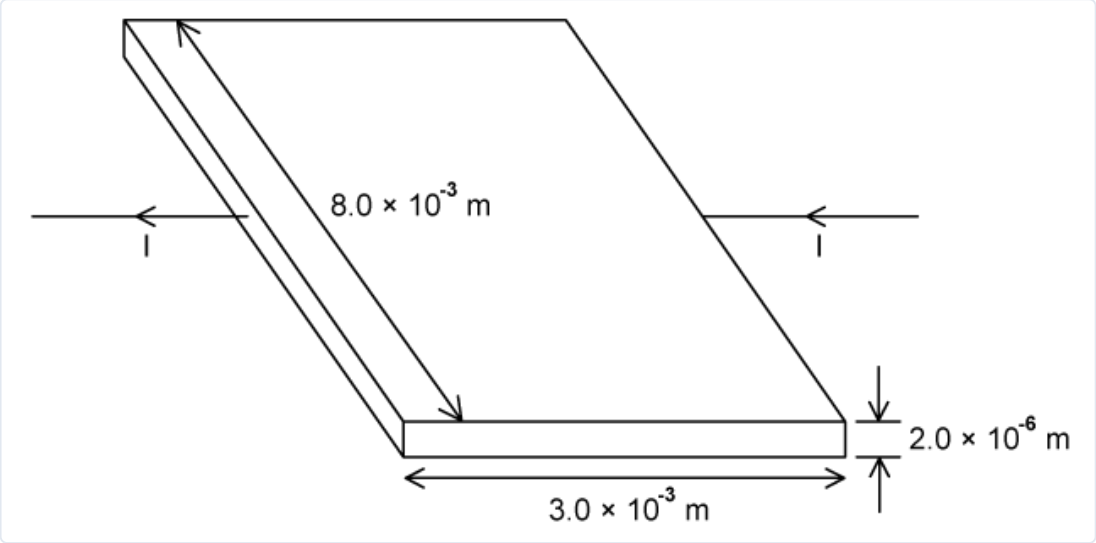


Fig. 1.1.

Calculate the current which passes through the carbon film in Fig. 1.1. for an applied voltage of 2.5 mV.

The resistivity of carbon = 4.0×10^{-5} Ω m

Part (b) [4 marks]

The applied voltage is kept constant, but the current is now directed through the carbon film as shown in Fig. 1.2.

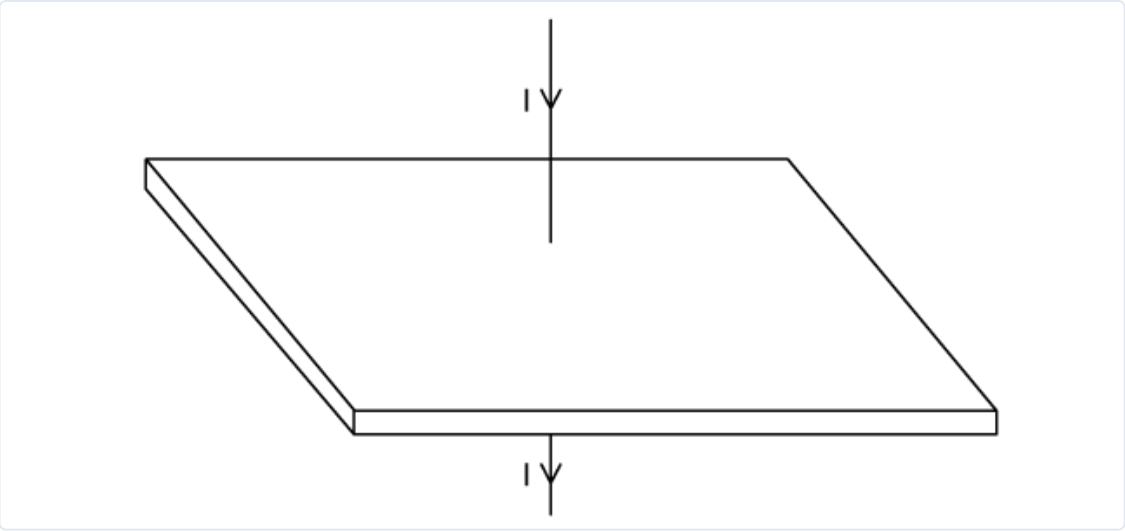


Fig. 1.2.

Show that the current is approximately a million times larger if it is directed through the carbon film as shown in Fig. 1.2.

Part (c) **[4 marks]**

A **tensile** force is applied to the carbon film as shown in Fig. 1.2 in a plane that is normal to the current.

Without performing any calculations, discuss how the resistance of the carbon film changes as a result of the applied tensile force.

State any assumptions you make in your answer.